

USING RESOURCES – CORROSION & RUSTING

KEY VOCABULARY

Corrosion: the gradual destruction of a metal by reaction.

Rusting: the corrosion of iron, needing oxygen and water.

Barrier method: painting/oiling to keep out air and water.

Sacrificial protection: a more reactive metal corrodes instead.

COMMON MISCONCEPTIONS

~~—Rusting needs only oxygen.~~

✓ Rusting needs BOTH oxygen and water.

~~—All metals rust.~~

✓ Only iron and steel rust; other metals corrode differently.

~~—Galvanising only works as a barrier.~~

✓ Zinc also gives sacrificial protection.

PREVENTING RUST

Rusting needs oxygen AND water.

Barrier: paint, oil, grease, plastic coating.

Galvanising: coat iron with zinc.

Sacrificial: a more reactive metal corrodes first.

Word equation: iron + oxygen + water → hydrated iron(III) oxide.

1

WHAT RUST NEEDS

Name the two substances needed for iron to rust. [2 marks]

2

BARRIER METHODS

Explain how painting prevents rusting. [2 marks]

3

SACRIFICIAL

Explain how sacrificial protection stops iron from corroding. [3 marks]

Refer to the reactivity series.

4

GALVANISING

State what is meant by galvanising. [1 mark]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Rusting needs ____ and ____; galvanising coats iron with zinc.

oxygen

water

zinc

carbon

6

WHY ZINC WORKS

Explain why zinc protects iron even if the coating is scratched. [2 marks]